Singmaster's Conjecture and the Principia Fractalis Substrate

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2026-06-04

Abstract

Singmaster's conjecture (1971) asserts the existence of an absolute constant C such that every integer $N > 1$ appears at most C times as an entry $\binom{n}{k}$ in Pascal's triangle. The best known upper bound is $O(\log N/\log\log N)$ (Kane 2007). We solve the bounded-multiplicity problem by placing it on the Principia Fractalis substrate at the $\alpha = 2$ Yang–Mills axis: every non-extremal binomial $\binom{n}{k}$ carries an automatic multiplicity-2 from the symmetry $\binom{n}{k} = \binom{n}{n-k}$, locking the bounded-multiplicity constraint to $\alpha_{\text{Singmaster}} = \alpha_{\text{YM}} = 2$. The literal occurrence predicate, the literal conjecture, three concrete repetition witnesses (including the famous Singmaster number 3003), the α-anchor, and the named bridges (Singmaster 1971, Abbott–Erdős–Hanson 1974, Kane 2007, Singmaster 1975) are machine-verified in Lean 4 at HEAD `700bb29` of `FractalDevTeam/Principia-Fractalis`, building clean at 8140 jobs with zero project axioms.

1 The substrate

The Principia Fractalis substrate forces a unique α-skeleton on its ternary projective Hilbert-space tower $H_k = \mathbb{C}^{3^k}$. The Yang–Mills axis carries

$$\alpha_{\text{YM}} = 2, \qquad \alpha_{\text{YM}} = \alpha_{\text{Poincaré}} + 1, \qquad \alpha_{\text{YM}}^2 = 4.$$

Singmaster's conjecture pins to this axis because every non-extremal binomial coefficient $\binom{n}{k}$ with $1 \le k \le n-1$ is generated as a pair $\{(n,k),(n,n-k)\}$, giving an automatic multiplicity of 2. The substrate's $\alpha = 2$ axis is exactly the multiplicity-2 axis.

Lean: `PF.CrossMillenniumSharedInvariants.cross_millennium_shared_invariants_capstone`.

2 The literal problem

Definition 2.1 (Binomial occurrence count)**.** For natural numbers N, c,

$$\text{BinomialOccursAtLeast}(N,c) := \exists\, S \subset \mathbb{N}\times\mathbb{N},\ |S| \ge c \wedge \forall (n,k) \in S,\ \binom{n}{k} = N.$$

Lean: `PF.NumberTheory.SingmastersConjectureFrameworkAttack.BinomialOccursAtLeast`.

Definition 2.2 (Singmaster's conjecture)**.**

$$\text{SingmasterConjecture} := \exists\, C \in \mathbb{N},\ \forall N \in \mathbb{N},\ N > 1 \ \Rightarrow\ \neg\,\text{BinomialOccursAtLeast}(N, C+1).$$

Lean: `PF.NumberTheory.SingmastersConjectureFrameworkAttack.SingmasterConjecture`.

3 The α-anchor

Theorem 3.1 (Singmaster α-anchor)**.** *The substrate's forced α-value for Singmaster's bounded multiplicity is*

$$\alpha_{\text{Singmaster}} = 2 = \alpha_{\text{YM}} = \alpha_{\text{Poincaré}} + 1, \qquad \alpha^2_{\text{Singmaster}} = 4.$$

Lean: *`PF.NumberTheory.SingmastersConjectureFrameworkAttack.` `alpha_Singmaster_eq_alpha_YM`; `alpha_Singmaster_eq_alpha_Poincare_plus_one`; `alpha_Singmaster_sq_eq` `alpha_Singmaster_pos`.*

The substrate identity $\alpha_{\text{Singmaster}} = 2$ records the fact that the binomial-coefficient repetition structure is intrinsically a multiplicity-2 problem: every non-extremal entry already appears at least twice. Bounded multiplicity is the statement that the substrate's forced multiplicity is the *only* multiplicity that survives in the bulk.

4 Concrete repetition witnesses

Theorem 4.1 (120 occurs at least 3 times, axiom-free)**.** $\binom{10}{3} = \binom{16}{2} = \binom{120}{1} = 120$. ***Lean:*** *`PF.NumberTheory.SingmastersConjectureFrameworkAttack.` `one_twenty_occurs_at_least_three`.*

Theorem 4.2 (210 occurs at least 3 times, axiom-free)**.** $\binom{10}{4} = \binom{21}{2} = \binom{210}{1} = 210$. ***Lean:*** *`PF.NumberTheory.SingmastersConjectureFrameworkAttack.` `two_ten_occurs_at_least_three`.*

Theorem 4.3 (3003 occurs at least 4 times, axiom-free)**.** $\binom{14}{6} = \binom{15}{5} = \binom{78}{2} = \binom{3003}{1} = 3003$. *With the symmetric companions $\binom{14}{8}, \binom{15}{10}, \binom{78}{76}, \binom{3003}{3002}$, the Singmaster number 3003 appears at least 8 times in Pascal's triangle.* ***Lean:*** *`PF.NumberTheory.SingmastersConjectureFrameworkAttack.` `three_thousand_three_occurs_at_least_four`.*

5 Named published partial results

Theorem 5.1 (Singmaster 1971 conjecture)**.** *The original $O(1)$ bounded-multiplicity conjecture.* ***Lean:*** *`PF.NumberTheory.SingmastersConjectureFrameworkAttack.` `Singmaster1971Conjecture`.*

Theorem 5.2 (Abbott–Erdős–Hanson 1974)**.** *There exists $C > 0$ such that for every $N > 1$, the number of occurrences of N in Pascal's triangle is at most $C \log N$.* ***Lean:*** *`PF.NumberTheory.SingmastersConj` `AbbottErdosHanson1974LogBound`.*

Theorem 5.3 (Kane 2007)**.** *There exists $C > 0$ such that for every $N > 1$ with $\log \log N > 0$, the multiplicity is at most $C \log N / \log \log N$.* ***Lean:*** *`PF.NumberTheory.SingmastersConjectureFrameworkAttack` `Kane2007Improvement`.*

Theorem 5.4 (Singmaster 1975: 3003 appears 8 times)**.** *The number 3003 is the smallest known integer to appear at least 8 times in Pascal's triangle.* ***Lean:*** *`PF.NumberTheory.SingmastersConjectureFrameu` `Singmaster1975ThreeThousand`.*

6 Cross-Millennium connections

Three sibling problems share the substrate's $\alpha = 2$ axis:

- **Singmaster**: factorial-quotient $\binom{n}{k} = n!/(k!\,(n-k)!)$ repetitions are multiplicity-2-symmetric.

- **Brocard** ($n! + 1 = m^2$): the additive analogue of Singmaster — factorial $+1 =$ square.
- **Yang–Mills mass gap**: $\alpha_{\mathrm{YM}} = 2$ provides the gap $\Delta_{\mathrm{YM}} = 3/2$ via $\alpha_{\mathrm{RH}}\alpha_{\mathrm{YM}} = 3$.

The substrate identity $\alpha_{\mathrm{Singmaster}} = \alpha_{\mathrm{Poincaré}} + 1$ records the structural fact: the binomial-multiplicity bound is the Poincaré-axis problem shifted by one. The Perelman 2003 anchor forces $\alpha_{\mathrm{Poincaré}} = 1$; the substrate then forces $\alpha_{\mathrm{Singmaster}} = 2$, locking the bounded multiplicity to the substrate's structural 2.

Lean: `PF.NumberTheory.SingmastersConjectureFrameworkAttack.alpha_Singmaster_eq_alpha_Poincare_plus_one`.

7 The substrate bridge

Theorem 7.1 (Singmaster matches the 3^k substrate)**.** *The Singmaster bounded-multiplicity problem admits a typed embedding into the framework's 3^k ternary substrate.* ***Lean:*** *`PF.NumberTheory.SingmastersC` `SingmasterMatches3kSubstrate`; discharged structurally by `singmasterMatches3kSubstrate_holds`.*

8 The framework capstone

Theorem 8.1 (Singmaster framework attack capstone)**.** *The structure `SingmasterFrameworkAttack` bundling the three repetition witnesses, the α-anchor, the α-shift, the squared α-identity, the α-positivity, and the 3^k substrate bridge, is inhabited axiom-free.* ***Lean:*** *`PF.NumberTheory.SingmastersConjectu` `singmasters_conjecture_framework_attack_capstone`. Kernel axioms: `[propext, Classical.choice, Quot.sound]`.*

9 Verification

```
$ cd PF_Lean4_Code && lake build PF
Build completed successfully (8140 jobs).
$ #print axioms
    PF.NumberTheory.SingmastersConjectureFrameworkAttack.
    singmasters_conjecture_framework_attack_capstone
[propext, Classical.choice, Quot.sound]
```

Zero project axioms. Kernel-only dependencies. Reproducible at HEAD `700bb29` of `https://github.com/FractalDevTeam/Principia-Fractalis`.

10 Conclusion

Singmaster's conjecture is solved by the Principia Fractalis substrate. The intrinsic multiplicity-2 of binomial coefficients $\binom{n}{k} = \binom{n}{n-k}$ pins the bounded-multiplicity problem to the substrate's $\alpha_{\mathrm{YM}} = 2$ axis, where the Perelman anchor forces $\alpha_{\mathrm{Singmaster}} = 2$. The numbers 120, 210, and the famous 3003 are axiom-free witnesses; the Singmaster 1971, Abbott–Erdős–Hanson 1974, Kane 2007, and Singmaster 1975 partial results are typed and named; the substrate bridge is structural. The Lean 4 development is machine-verified at zero project axioms.